

Regression Models

Data Mining Project Report

February 5, 2026

Abstract

This report presents the development and evaluation of regression models for predicting laptop prices in the Algerian market. Using a dataset of over 15,000 listings from Ouedkniss, we compare linear and tree-based approaches. Our best model, XGBoost, achieves an R^2 of 0.827 on unseen test data, demonstrating that hardware performance benchmarks, particularly CPU and GPU scores, are the primary determinants of laptop pricing.

1 Introduction

Laptop pricing in informal marketplaces lacks standardization. Sellers set prices based on intuition rather than objective hardware value, creating information asymmetry between buyers and sellers.

This study addresses two questions:

1. Can laptop prices be accurately predicted from hardware specifications?
2. Which features most strongly influence pricing decisions?

2 Data Overview

The dataset contains 15,517 laptop listings with 15 features after preprocessing. The target variable is `price_preview` (in Algerian Dinars), which was log-transformed to handle right-skewness.

2.1 Feature Set

The final features used for modeling include:

- **Performance metrics:** `cpu_mark`, `gpu_g3d_mark`, `gpu_g2d_mark`, `cores`, `tdp`, `gpu_tdp`
- **Hardware specs:** `RAM_SIZE`, `SSD_SIZE`, `HDD_SIZE`
- **Display:** `SCREEN_RESOLUTION_ENC`
- **Categorical:** `model_family` (brand tier), `spec_Etat` (condition)

2.2 Data Splitting

The data was split using stratified sampling based on price bins to ensure representative distributions across all sets:

- Training: 9,310 samples (60%)
- Validation: 3,103 samples (20%)
- Test: 3,104 samples (20%)

3 Methodology

3.1 Feature Selection

Sequential Backward Elimination was applied to reduce dimensionality from 14 to 10 features. This process removed low-importance features such as `match_score` and `HDD_SIZE`, which contributed noise rather than signal.

3.2 Models Evaluated

Six regression models were compared:

1. Linear Regression (baseline)
2. Ridge Regression (L_2 regularization)
3. Lasso Regression (L_1 regularization)
4. Random Forest
5. XGBoost
6. LightGBM

Tree-based models (4–6) were tuned using grid search with 5-fold cross-validation. Hyperparameters optimized for XGBoost included `learning_rate`, `max_depth`, `n_estimators`, and regularization terms.

4 Results

4.1 Model Comparison

Table 1: Model Performance on Test Set

Model	R^2	RMSE (DZD)	MAE (DZD)
XGBoost	0.8271	41,231	21,070
LightGBM	0.8144	42,714	21,596
Random Forest	0.8088	43,362	21,747
Ridge Regression	0.4801	71,497	32,318
Lasso Regression	0.4806	71,462	32,337
Linear Regression	0.4730	71,981	32,386

XGBoost achieved the highest R^2 of 0.827, explaining approximately 83% of the variance in laptop prices. The Mean Absolute Error of 21,070 DZD indicates that, on average, predictions deviate by about 21,000 DZD from actual prices.

4.2 Linear vs. Tree-Based Models

Linear models plateaued at $R^2 \approx 0.48$, while tree-based models reached $R^2 > 0.80$. This 72% improvement demonstrates that laptop pricing follows non-linear patterns. For example, the value of additional RAM depends on the CPU tier and brand context—relationships that linear models cannot capture.

4.3 Feature Importance

Table 2 shows the relative importance of features in the XGBoost model.

Table 2: Feature Importance (XGBoost)

Feature	Importance
cpu_mark	High
gpu_g3d_mark	High
RAM_SIZE	Medium
SSD_SIZE	Medium
model_family	Medium
SCREEN_RESOLUTION_ENC	Low
spec_Etat	Low

CPU and GPU performance benchmarks dominate pricing, followed by memory and storage capacity. Brand tier (`model_family`) captures premiums not explained by raw specifications.

4.4 Residual Analysis

Residuals were examined to assess model validity:

- **Bias:** Residuals are centered around zero, indicating no systematic over- or under-prediction.
- **Variance:** Slight heteroscedasticity was observed at higher price points. Premium laptops (>200,000 DZD) exhibit greater prediction variance, likely due to intangible factors (build quality, brand reputation) not captured by hardware specs alone.

5 Discussion

5.1 Key Findings

1. **Performance benchmarks matter more than marketing labels.** The `cpu_mark` score has stronger predictive power than CPU generation. A 10th-generation Core i7 with a high benchmark score commands higher prices than a 13th-generation Core i3 with a lower score.
2. **Storage type is decisive.** SSDs carry a significant price premium over HDDs. Laptops with HDD-only storage are predominantly older models (pre-2020) and are valued lower regardless of other specs.
3. **GPU matters almost as much as CPU.** The `gpu_g3d_mark` feature ranks second in importance, reflecting the market premium for gaming and graphics workloads.
4. **Condition labeling is mistrusted.** Despite expectations, the “Never Used” condition label provides only a modest price premium (+12% vs. expected +25%). This suggests buyer skepticism toward unverified claims.

5.2 Limitations

- Intangible factors (e.g., screen quality, keyboard feel) are not represented in the feature set.
- Temporal effects (price depreciation over time) were not modeled.
- The dataset is specific to the Algerian market and may not generalize to other regions.

6 Conclusion

This study demonstrates that laptop prices can be predicted with high accuracy ($R^2 = 0.827$) using hardware performance benchmarks. The success of tree-based models over linear approaches confirms that pricing follows complex, non-linear patterns driven primarily by CPU and GPU performance.

The resulting model has practical value for:

- **Buyers:** Identifying overpriced listings and negotiating fair prices.
- **Sellers:** Setting competitive prices based on objective hardware value.
- **Market analysis:** Understanding which features drive value in the laptop market.

Future work could incorporate time-series analysis to capture depreciation trends and expand the model to include tablets and desktops.

Appendix: Model Configuration

XGBoost Hyperparameters (Best Configuration):

```
learning_rate: 0.1
max_depth: 7
n_estimators: 100
reg_alpha: 1.0
```

Feature Selection: 10 features retained from 14 candidates using Sequential Backward Elimination with cross-validated R^2 as the scoring metric.